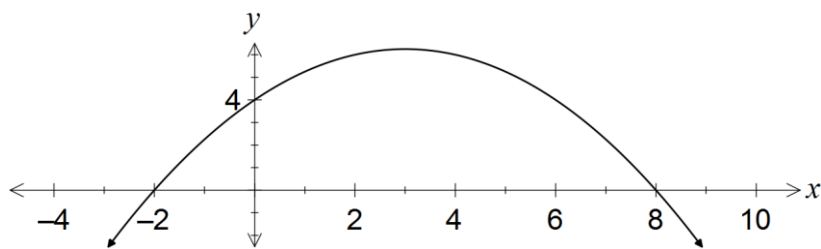


Question 1
(7 marks)

- (a) Part of the graph of $y = ax^2 + bx + 4$ is shown below.



Determine the values of the coefficients a and b .

(3 marks)

Solution
$y = a(x + 2)(x - 8)$ $(0, 4) \Rightarrow 4 = a(2)(-8) \Rightarrow a = -\frac{1}{4}$ $y = -\frac{1}{4}(x^2 - 6x - 16)$ $= -\frac{1}{4}x^2 + \frac{3}{2}x + 4 \Rightarrow a = -\frac{1}{4}, b = \frac{3}{2}$
Specific behaviours
✓ uses roots to express in factored form ✓ uses y-intercept to find a ✓ expands and states b

- (b) A quadratic has equation $y = x^2 - 6x + 2$. Determine

- (i) the coordinates of its turning point.

(2 marks)

Solution
$x^2 - 6x + 2 = (x - 3)^2 - 3^2 + 2$ $= (x - 3)^2 - 7$ At $(3, -7)$
Specific behaviours
✓ completes square, or uses $x = -b/2a$ ✓ states coordinates

- (ii) the exact values of the zeros of the quadratic.

(2 marks)

Solution
$(x - 3)^2 - 7 = 0$ $x - 3 = \pm\sqrt{7}$ $x = 3 \pm \sqrt{7}$
Specific behaviours
✓ uses quadratic formula or completes square ✓ states both roots in exact form

- (c) Is it possible to bend a 12 cm length of wire to form the perpendicular sides of a right angled triangle with area 20cm?
(4 marks)

Solution
Height = x , Base = $12-x$ Area: $20 = \frac{1}{2}x(12 - x)$ $40 = \frac{1}{2}x(12 - x)$ $12x - x^2 - 40 = 0$ $x^2 - 12x + 40 = 0$ Discriminant = $(-12)^2 - 4(1)(40)$ $= -16$ which is < 0 <i>There are no real solutions, indicating this situation is impossible.</i>
Specific behaviours
✓ Use of x and $12 - x$ correctly. ✓ Substituting into area of a triangle formula ✓ Correct general formula ✓ Use of discriminant to indicate no real solutions. Note: 1 mark if they indicate that they would need two numbers which add to 12 and multiply to 40, 1 mark if they try some values to show that it is not possible, 1 mark if they set out a table in an orderly manner and reach the maximum of 6x6 giving 36 (ie max area is 18sq m) and 1 mark for demonstrating that this is the maximum by extending the table etc. Basically use your professional judgement and (generously) allocate a mark out of 4 accordingly.

Question 2**(8 marks)**

(a) A circle of radius 5 has its centre at (6, -4).

(i) Determine the equation of this circle.

(2 marks)

Solution
$(x - 6)^2 + (y + 4)^2 = 25$
Specific behaviours
✓ uses standard circle form with correct radius ✓ correct equation

(ii) State, with justification, whether the point (9, -8) lies on the circle.

(1 mark)

Solution
$(9 - 6)^2 + (-8 + 4)^2 = 9 + 16 = 25 \Rightarrow$ Does lie on circle
Specific behaviours
✓ substitutes point into equation from (a) and interprets

(b) Determine the centre and radius of the circle with equation $x^2 + y^2 - 4x + 6y + 9 = 0$ **(3 marks)**

Solution
$(x - 2)^2 - 4 + (y + 3)^2 - 9 + 9 = 0$ $(x - 2)^2 + (y + 3)^2 = 4 = 2^2$ Hence centre at (2, -3) and radius 2
Specific behaviours
✓ factors x terms ✓ factors y terms ✓ states centre and radius

(c) Find the equation of the curve drawn below.

(3 marks)

Solution
$y = k\sqrt{x + b} + c$ $y = 2\sqrt{x + 3} - 2$
Specific behaviours
✓ $a = 3$ ✓ $k = 2$ ✓ $c = -2$

Question 3 (1.1.14)**(2, 2, 2 = 6 marks)**

A rectangular hyperbola has asymptotes with equation $x = -2$ and $y = 4$.

a) Write two possible equations for this function

Solution
$y = \frac{a}{x+2} + 4$ so a could be any number eg $y = \frac{1}{x+2} + 4$ and $y = \frac{-1}{x+2} + 4$
Specific behaviours
✓✓ two possible equations

b) Write the equation of this function if it has a y-intercept at (0,5)

Solution
$5 = \frac{a}{0+2} + 4$ so $a=2$
Specific behaviours
✓ substitutes correctly into equation ✓ $a=2$

c) Write the equation of this function if it passes through the point (3,5)

Solution
$5 = \frac{a}{3+2} + 4$ so $a=5$ therefore $y = \frac{5}{x+2} + 4$
Specific behaviours
✓ substitutes correctly into equation ✓ states equation

Question 4 (1.1.24)**(1, 2, 1, 2 = 6 marks)**

a) Given $f(x) = x^2 - 2x$

i) What type of correspondence does f show? Circle one of the following.

Many-to-one

One-to-many

One-to-one

Specific behaviours
✓ Many to one

ii) If the domain of f is $f(x) \in R, -4 \leq x \leq 5$, find the range of f .

Specific behaviours
✓✓ $-1 \leq y \leq 24$

b) Given $y = 2 + \sqrt{4 - x^2}$

i) What is the largest possible value of y .

Specific behaviours
✓ $y = 4$

ii) Determine the domain and range.

Specific behaviours
✓ $-2 \leq x \leq 2$ ✓ $2 \leq y \leq 4$

Question 5 (1.1.24)**(1, 1, 2, 2 = 6 marks)**Suppose $G(x) = \frac{2x-3}{x-4}$.a) Evaluate $G(2)$

Solution
✓ $-\frac{1}{2}$

b) Find a value of x such that $G(x)$ does not exist.

Solution
✓ $x = 4$

c) Find $G(x + 2)$ in simplest form.

Solution
$g(x + 2) = (2(x + 2) - 3)/((x + 2) - 4)$ $g(x + 2) = \frac{2x + 1}{x - 2}$
Specific behaviours
✓ Substitute correctly ✓ Answer

d) Find x such that $G(x) = -3$.

Solution
$-3 = \frac{2x - 3}{x - 4}$ $x = 3$
Specific behaviours
✓ Sets equation up correctly ✓ Answer